USN. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18MTFT232]

SECOND SEMESTER M.TECH EXAMINATION, JUNE – 2019

ENZYME AND FERMENTATION TECHNOLOGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
ii. Draw neat sketches wherever necessary.
iii. Make suitable assumptions wherever necessary.

1.	**a.**	With reference to the nature of reaction catalysed, breakdown the classification scheme of enzymes.	**10**
	b.	Compare the mode of action of Lyases and Hydrolases with appropriate examples.	**10**
2.	**a.**	Discuss the mechanistic differences between the increase in the rate of a chemical reaction brought about by an enzyme and the heating of the reaction mixture.	**10**
	b.	List out various inhibitions mechanisms of enzyme reactions with examples.	**10**
3.	**a.**	Analyze the mode of action and applications of Pectinase in fruit industry, with an example.	**08**
	b.	Prioritize the scope of the following: **i.** Xanthine oxidase **ii.** Catalase and its application	**12**
		OR	
4.	**a.**	Formulate the objectives of immobilization of enzymes with suitable examples.	**08**
	b.	Elaborate the scope and applications of catalases and peroxidases in food industry.	**12**
5.	**a.**	With the help of a neat schematic diagram illustrate the continuous stirred tank bioreactor.	**10**
	b.	"Enhanced efficiency by automation of bioreactors" critically comment on the statement.	**10**
		OR	
6.	**a.**	Compare the advantages and disadvantages of fluidized bed bioreactor over air lift bioreactor.	**10**
	b.	Inspect the major challenges of bioreactor design and operations with a case study.	**10**
7.	**a.**	Infer various sources of carbon and nitrogen in media formulation.	**08**
	b.	What is the role of fermentation media? Compare synthetic media over crude media.	**12**
		OR	
8.	**a.**	Compare batch and continuous sterilization process with a special emphasis on their merits and demerits.	**08**
	b.	Substantiate "Fermentation media must satisfy all the nutritional requirements of the microorganism and fulfil the technical objectives of the process".	**12**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112